



MEGN261: Spring 2026

EES Chevron Project

Team 122 (NED)

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Introduction/Background

In today's power generation industry, improving the efficiency of combined-cycle natural gas power plants is very important for many different reasons in industry, specifically from both an economic and environmental perspective. As electricity and power demand continue to increase, power companies have to maximize energy output while minimizing the costs to operate and produce this energy. Even small increases in efficiencies can turn into significant reductions in things like fuel usage and spending. Because of this, using and upgrading and optimizing already existing power plants is often much more practical and financially beneficial than creating and investing in entire new facilities.

Chevron has tasked our engineering team (Team NED) with evaluating the performance of one of their combined-cycle power plants and identifying modifications that could improve its overall efficiency and net power output. The baseline plant, which we analyzed in EES, worked at an overall thermal efficiency of approximately 33.13% and generated a net power output of 42.47 MW. While several individual parts and states had higher performance, including turbine isentropic efficiencies above 92% and compressor efficiency near 83.34%, the system still had thermal and compressor losses that lowered the overall performance and output of the plant.

The purpose of this project was to move beyond simple baseline calculations, like in Part 2 of this project, and perform a more detailed analysis of these potential plant improvements. Using EES, our team created and solved for the states in the existing plant, and evaluated four separate upgrades: an evaporative cooling system, an OEM turbine upgrade allowing a turbine inlet temperature of 2100 °F, a higher-efficiency inlet air filter, and a dual-stage compressor with intercooling. Each mod and its values were solved on its own to determine its effects on cycle efficiency, net power output, compressor work and if it would be beneficial in its overall practicality.

This project highlights the importance of analysis in real world energy systems. Engineers like ourselves in the future must not only understand the math behind our work, but also evaluate things like economic feasibility, limitations and constraints and real-world factors of use. By combining thermodynamic mathematics with engineering assumptions, this analysis gives Chevron a data-driven framework for figuring out future plant upgrades that maximize performance while being realistic.

Objectives

The primary objective of this project was to model, analyze and improve the performance of the power plant using thermodynamic analysis in EES. The project focused on evaluating multiple different plant upgrades and figuring out how each modification affected overall plant efficiency, net power generation, compressor work and operational feasibility. The specific objectives of the project are listed below.

1. Evaporative Cooler Before Air Filter Inlet

The first modification was an evaporative cooler placed upstream of the compressor inlet. The purpose of this modification was to reduce compressor inlet temperature, increase inlet air density and decrease compressor work while maintaining a constant volumetric airflow rate. The analysis also includes a financial aspect where we have to determine the ROI of a mock price of installation, energy and price per hour.

2. OEM Turbine Upgrades

The second modification was doing OEM upgrades to turbine parts that would safely increase the turbine inlet temperature to 2100 °F. This increase of the turbine inlet temperature would raise the maximum Brayton cycle temperature, which in turn increases turbine specific work and overall thermal efficiency. This modification and analysis also contained questions on what impacts do the upgrades have on the cycle and on any potential concerns that may be present with these modifications.

3. Finer Air Filter Before the Compressor

The third objective was to examine the effects of installing a finer inlet air filtration system before the compressor. Although cleaner inlet air could potentially improve compressor performance and reduce blade fouling, the filter also introduced an additional 2 psi pressure drop. The goal was to determine the compressor efficiency increase required to offset the thermodynamic losses caused by the pressure drop and evaluate whether the modification was practical.

4. Dual Stage Compressor With Intercooling

The final modification was one where our team completed research and analysis on an additional plant upgrade and modification. Our team decided on attempting to use a dual-stage compressor with intercooling. This modification was designed to reduce total compressor work by splitting compression into two stages and cooling the air between stages, allowing similar air pressures into the combustor (to baseline) with lower specific volume and temperature, thus increasing air molecule density. The objective was to determine how intercooling affected compressor work, cycle efficiency, turbine performance, and overall net power generation.

Methodology

The Chevron combined-cycle power plant and all upgrades were modeled using EES which allowed us to use state properties and energy and mass balances to solve for values throughout the cycle, making it possible to calculate the changes and modifications done to the entire system. The software was very useful for doing the calculations which involved compressors, turbines, combustion processes, heat exchangers and psychrometric relationships.

The baseline plant was first developed and validated using the provided plant data and P&ID. Thermodynamic properties including temperature, pressure, enthalpy, entropy and mass

flow rates were calculated at each state point in the cycle. Arrays were used in EES to organize and show state data and generate T-s diagrams for both the baseline and upgraded systems.

Assumptions made to simplify the analysis:

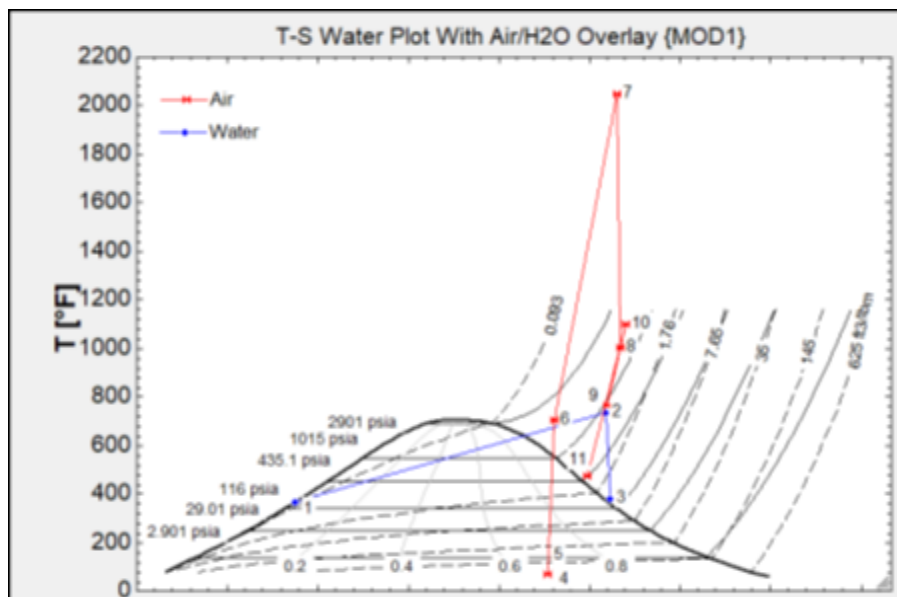
- The plant operates at steady-state conditions
- Changes in kinetic and potential energy are negligible
- Compressor and turbine efficiencies remain constant unless modified
- Air at the compressor inlet is humid air (AirH2O)
- High-temperature regions are dry air (Air)
- The evaporative cooler maintains constant volumetric airflow through the compressor
- The intercooler operates under ideal or near-perfect cooling assumptions

One of the challenges we had during the process was to keep certain states stable and realistic throughout the cycle. Multiple tries of debugging the code equations were required to fix incorrect or impossible numbers in the data.

Results

MOD 1: Evaporative Cooler

For our first modification, we modeled an evaporative cooler before the compressor inlet to reduce the inlet air temperature and increase air density. Since we assumed that the volumetric airflow through the compressor was constant, the cooler inlet air increased the mass flow rate that entered the Brayton cycle while it also reduced the specific work required by the compressor. According to psychrometric cooling principles, evaporative cooling lowers dry-bulb temperature while increasing humidity which leads to improving compressor performance in hot operating conditions [1]. The EES analysis showed that this modification increased the overall thermal efficiency from 33.13% to 34.64% and increased net power output from 42.47 MW to 43.73 MW. Although this system produced the largest efficiency increase of all modifications, it also introduces additional installation and operational costs associated with water usage and cooling equipment [1]. We also completed the ROI analysis of the provided installation example, which showed us that it would take 165.3 days to break even on the investment.



Results and Changes to EES Code:

Unit Settings: Eng F psia mass deg			
Capital _{cost} = 1000000 [\$]	Cost _{per,hr} = 100 [\$ /hr]	$\dot{Q}_{W,MW}$ = 1.258 [MW]	h_{10} = 385.7 [Btu/lb]
h_{11} = 224.2 [Btu/lb]	h_4 = 126.4 [Btu/lb]	h_5 = 126.4 [Btu/lb]	$h_{5,raw}$ = 126.6 [Btu/lb]
h_6 = 282.5 [Btu/lb]	h_{6s} = 256.5 [Btu/lb]	$h_{6,raw}$ = 274.7 [Btu/lb]	h_7 = 648.6 [Btu/lb]
$h_{7,raw}$ = 648.6 [Btu/lb]	h_8 = 359.2 [Btu/lb]	h_{8s} = 342.6 [Btu/lb]	$h_{8,raw}$ = 367.9 [Btu/lb]
h_9 = 297.2 [Btu/lb]	$h_{feedwater}$ = 342.5 [Btu/lb]	$h_{st,in}$ = 1356 [Btu/lb]	$h_{st,out}$ = 1205 [Btu/lb]
$h_{st,out,s}$ = 1193 [Btu/lb]	$\dot{m}_{air}$ = 263.1 [lb/s]	$\dot{m}_{air,base}$ = 262.8 [lb/s]	$\dot{m}_{H2O}$ = 41.95 [lb/s]
$\dot{m}_{steam,b,hr}$ = 151016 [lb/hr]	Net _{profit,per,hr} = 252.1 [\$ /hr]	n_{comp} = 0.8334	n_{cooler} = 0.9
n_{cycle} = 0.3464	$n_{t,GT}$ = 0.9457	$n_{t,st}$ = 0.9256	Payback _{days} = 165.3 [days]
Payback _{hours} = 3966 [hrs]	Payback _{years} = 0.4528 [yrs]	price _{MWh} = 280 [\$ /MWh]	P_{10} = 15.7 [psia]
P_{11} = 14.7 [psia]	P_4 = 13.7 [psia]	P_5 = 13.7 [psia]	P_6 = 162.6 [psia]
P_7 = 162.6 [psia]	P_8 = 15.7 [psia]	P_9 = 15.7 [psia]	P_{atm} = 14.7 [psia]
$P_{st,in}$ = 869.7 [psia]	$P_{st,out}$ = 171.2 [psia]	$\dot{Q}_{comb}$ = 96343 [Btu/s]	$\dot{Q}_{DB}$ = 23288 [Btu/s]
$\dot{Q}_{loss}$ = 16297 [Btu/s]	RelHum _{ambient} = 0.2	Revenue _{per,hr} = 352.1 [Rev/hr]	s_{10} = 1.621 [Btu/lb-R]
s_{11} = 1.494 [Btu/lb-R]	s_4 = 1.362 [Btu/lb-R]	s_5 = 1.362 [Btu/lb-R]	s_6 = 1.385 [Btu/lb-R]
s_{6s} = 1.362 [Btu/lb-R]	s_7 = 1.592 [Btu/lb-R]	s_8 = 1.604 [Btu/lb-R]	s_{8s} = 1.592 [Btu/lb-R]
s_9 = 1.558 [Btu/lb-R]	$s_{feedwater}$ = 0.5256 [Btu/lb-R]	$s_{st,in}$ = 1.554 [Btu/lb-R]	$s_{st,out}$ = 1.569 [Btu/lb-R]
$s_{st,out,s}$ = 1.554 [Btu/lb-R]	T_{10} = 1102 [F]	T_{11} = 471.9 [F]	T_4 = 68.86 [F]
T_5 = 68.86 [F]	T_6 = 705 [F]	T_7 = 2050 [F]	T_8 = 1002 [F]
T_9 = 763 [F]	$T_{ambient}$ = 95 [F]	$T_{feedwater}$ = 368.7 [F]	$T_{st,in}$ = 735 [F]
$T_{st,out}$ = 381.5 [F]	T_{wb} = 65.96 [F]	Vol_{flow} = 3761 [ft ³ /s]	$v_{5,base}$ = 14.31 [ft ³ /s]
$v_{5,new}$ = 14.29 [ft ³ /s]	$\dot{W}_{comp}$ = 41065 [Btu/s]	$\dot{W}_{GT}$ = 76164 [Btu/s]	$\dot{W}_{GT,Btu,base}$ = 34861 [Btu/s]
$\dot{W}_{GT,mw,base}$ = 36.78 [MW]	$\dot{W}_{net}$ = 43.73 [MW]	$\dot{W}_{net,Btu}$ = 41446 [Btu/s]	$\dot{W}_{ST}$ = 6347 [Btu/s]
$\dot{W}_{net,base}$ = 42.47 [MW]	$\dot{W}_{net,mod1}$ = 43.73 [MW]		

--MOD ONE - AMBIENT CONDITIONS ASSUMPTIONS--

$T_{ambient}$ = 95 [F]
 $RelHum_{ambient}$ = 0.20
 n_{cooler} = 0.90
 P_{atm} = 14.7 [psia]

"FOUND EFFICIENCIES"

n_{comp} = 0.8334
 $n_{t,GT}$ = 0.9457
 $n_{t,st}$ = 0.9256

"Isentropic Relations"

$s_5 = s_{6s}$
 $s_7 = s_{8s}$
 $s_{st,in} = s_{st,out,s}$

"RAW STATE ENTHALPIES"

$h_{5,raw} = \text{enthalpy}(\text{Air}, T=69.59)$
 $h_{6,raw} = \text{enthalpy}(\text{Air}, T=674.22)$
 $h_{7,raw} = \text{enthalpy}(\text{Air}, T=2050)$
 $h_{8,raw} = \text{enthalpy}(\text{Air}, T=1034.88)$

"STATE 5 - FILTER"

"AirH2O to find the tempe drop from the cooler"

$T_{wb} = \text{wetbulb}(\text{AirH2O}, T=T_{ambient}, P=P_{atm}, R=RelHum_{ambient})$
 $T_5 = T_{ambient} - n_{cooler} * (T_{ambient} - T_{wb})$
 $P_5 = (14.7 - 1)$
 $h_5 = \text{enthalpy}(\text{Air}, T=T_5)$
 $s_5 = \text{entropy}(\text{Air}, T=T_5, P=P_5)$

```

"Volumetric Flow Calcs - Keeping Vol Flow Constant"
v_5_base = volume(Air, T=69.59, P=(14.7 - 1)) "Calculated after the filter"
Vol_flow = m_dot_air_base * v_5_base

v_5_new = volume(Air, T=T_5, P=P_5)
m_dot_air = Vol_flow / v_5_new

"STEAM"
P_st_in = 854.96 + 14.7
P_st_out = 156.46 + 14.7
T_st_in = 735.02
T_feedwater = 368.71

h_feedwater = enthalpy(Steam, T=T_feedwater, P=P_st_in)
s_feedwater = entropy(Steam, T=T_feedwater, P=P_st_in)
h_st_in = enthalpy(Steam, T=T_st_in, P=P_st_in)
s_st_in = entropy(Steam, T=T_st_in, P=P_st_in)
h_st_out_s = enthalpy(Steam, s=s_st_in, P=P_st_out)
h_st_out = h_st_in - n_t_st*(h_st_in - h_st_out_s)
T_st_out = temperature(Steam, h=h_st_out, P=P_st_out)
s_st_out = entropy(Steam, h=h_st_out, P=P_st_out)

m_dot_H2O = (m_dot_air*(h_10 - h_11)) / (h_st_in - h_feedwater)

```

```

"--- ROI ANALYSIS ---"

"MOD 1 net power (MW)"
W_net_mod1 = W_dot_net

"Baseline Power Reference"
W_net_base = 42.47 [MW]

"Power increase due to cooler"
delta_W_MW = W_net_mod1 - W_net_base

"Electricity price"
price_MWh = 0.28*1000 [$ /MWh]

"Additional revenue per hour"
Revenue_per_hr = delta_W_MW * price_MWh

"Operating cost of cooler"
Cost_per_hr = 100 [$ /hr]

"Net profit per hour"
Net_profit_per_hr = Revenue_per_hr - Cost_per_hr

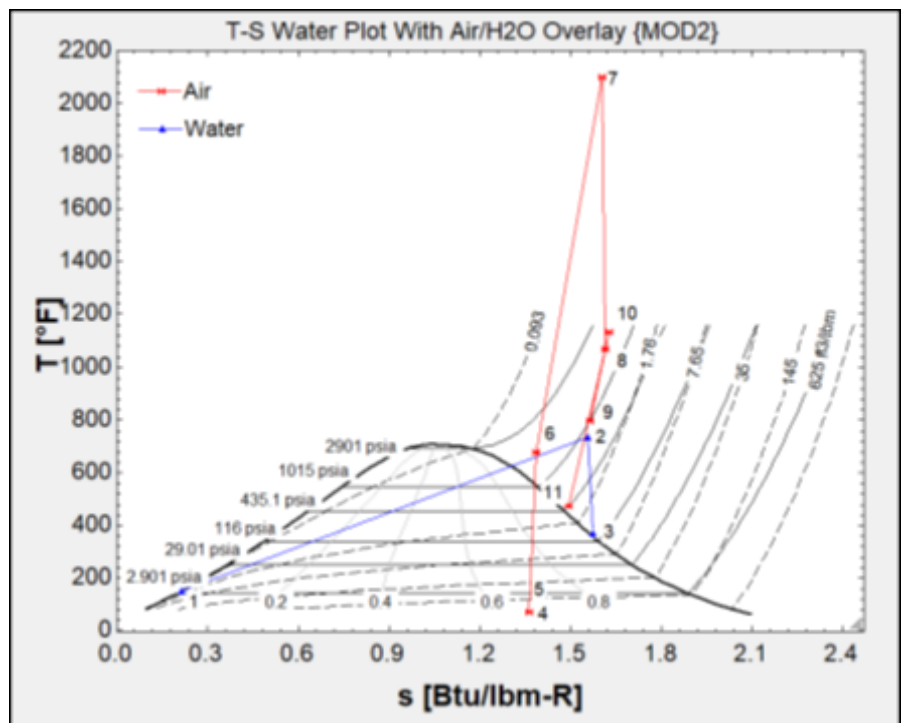
"Installation cost"
Capital_cost = 1000000 [$]

"Payback time"
Payback_hours = Capital_cost / Net_profit_per_hr
Payback_days = Payback_hours / 24 [hr/day]
Payback_years = Payback_days / 365 [day/yr]

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MOD 2: OEM Upgrades

For our second modification, we upgraded the gas turbine using OEM turbine upgrades that could increase the turbine inlet temperature to 2100 °F. Increasing the maximum Brayton cycle temperature raises the theoretical cycle efficiency while also increasing the specific work extracted by the turbine [2]. For this analysis, the turbine isentropic efficiency was assumed to remain constant relative to the baseline case. The hotter combustion gases ended up increasing turbine shaft work and also did raise the turbine exhaust temperature which improved the effectiveness of the downstream HRSG and steam cycle. The EES analysis showed that this modification increased thermal efficiency from 33.13% to 33.57% and increased net power output from 42.47 MW to 44.38 MW. Modern OEM upgrades use advanced blade cooling systems, thermal barrier coatings, and high-temperature materials to safely operate at elevated firing temperatures [2].



Results and Changes to EES Code:

Unit Settings: Eng F psia mass deg			
$h_1 = 120.2$ [Btu/lb]	$h_{10} = 394$ [Btu/lb]	$h_{11} = 224.2$ [Btu/lb]	$h_2 = 1357$ [Btu/lb]
$h_3 = 1200$ [Btu/lb]	$h_{3a} = 1187$ [Btu/lb]	$h_4 = 126.6$ [Btu/lb]	$h_5 = 126.6$ [Btu/lb]
$h_{5,raw} = 126.6$ [Btu/lb]	$h_6 = 274.7$ [Btu/lb]	$h_{6a} = 250$ [Btu/lb]	$h_{6,raw} = 274.7$ [Btu/lb]
$h_7 = 663$ [Btu/lb]	$h_{7,raw} = 648.6$ [Btu/lb]	$h_8 = 376.4$ [Btu/lb]	$h_{8a} = 359.9$ [Btu/lb]
$h_{8,raw} = 367.9$ [Btu/lb]	$h_9 = 305.5$ [Btu/lb]	$\dot{m}_1 = 36.08$ [lb/s]	$\dot{m}_{10} = 262.8$ [lb/s]
$\dot{m}_{11} = 262.8$ [lb/s]	$\dot{m}_2 = 36.08$ [lb/s]	$\dot{m}_3 = 36.08$ [lb/s]	$\dot{m}_4 = 262.8$ [lb/s]
$\dot{m}_5 = 262.8$ [lb/s]	$\dot{m}_6 = 262.8$ [lb/s]	$\dot{m}_7 = 262.8$ [lb/s]	$\dot{m}_8 = 262.8$ [lb/s]
$\dot{m}_9 = 262.8$ [lb/s]	$\dot{m}_{air} = 262.8$ [lb/s]	$\dot{m}_{H_2O} = 36.08$ [lb/s]	$\dot{m}_{steam,b,hr} = 129899$ [lb/hr]
$\eta_{comp} = 0.8334$	$\eta_{cycle} = 0.3357$	$\eta_{GT} = 0.9457$	$\eta_{t,st} = 0.9256$
$P_1 = 905$ [psia]	$P_{10} = 15.7$ [psia]	$P_{11} = 14.7$ [psia]	$P_2 = 855$ [psia]
$P_3 = 156.5$ [psia]	$P_4 = 13.7$ [psia]	$P_5 = 13.7$ [psia]	$P_6 = 147.9$ [psia]
$P_7 = 147.9$ [psia]	$P_8 = 15.7$ [psia]	$P_9 = 15.7$ [psia]	$\dot{Q}_{comb} = 102026$ [Btu/s]
$\dot{Q}_{DB} = 23256$ [Btu/s]	$\dot{Q}_{loss} = 18633$ [Btu/s]	$s_1 = 0.2142$ [Btu/lb-R]	$s_{10} = 1.627$ [Btu/lb-R]
$s_{11} = 1.494$ [Btu/lb-R]	$s_2 = 1.557$ [Btu/lb-R]	$s_3 = 1.572$ [Btu/lb-R]	$s_{3a} = 1.557$ [Btu/lb-R]
$s_4 = 1.362$ [Btu/lb-R]	$s_5 = 1.362$ [Btu/lb-R]	$s_6 = 1.385$ [Btu/lb-R]	$s_{6a} = 1.362$ [Btu/lb-R]
$s_7 = 1.604$ [Btu/lb-R]	$s_8 = 1.615$ [Btu/lb-R]	$s_{8a} = 1.604$ [Btu/lb-R]	$s_9 = 1.564$ [Btu/lb-R]
$T_1 = 150$ [F]	$T_{10} = 1133$ [F]	$T_{11} = 471.9$ [F]	$T_2 = 735$ [F]
$T_3 = 368.7$ [F]	$T_4 = 69.59$ [F]	$T_5 = 69.59$ [F]	$T_6 = 674.3$ [F]
$T_7 = 2100$ [F]	$T_8 = 1067$ [F]	$T_9 = 795.2$ [F]	$\dot{W}_{comp} = 38914$ [Btu/s]
$\dot{W}_{GAS,mw,base} = 36.78$	$\dot{W}_{GT} = 75304$ [Btu/s]	$\dot{W}_{GT,MW} = 79.45$ [MW]	$\dot{W}_{net} = 44.38$ [MW]
$\dot{W}_{raw,GAS} = 34861$ [Btu/s]	$\dot{W}_{ST} = 5671$ [Btu/s]		

"State 7 - After Combustor 1"

$T_7=2100$ {MOD 2 CHANGE: OEM Upgrade 2100F}
 $P_7=P_6$

"State 9 - After Duct"

$\{T_9=763\}$
 "1023.88-763 = 271.88" {MOD 2}
 $T_9 = T_8 - 271.88$ {MOD 2}
 $P_9=(14.7 + 1)$

"Gas Turbine"

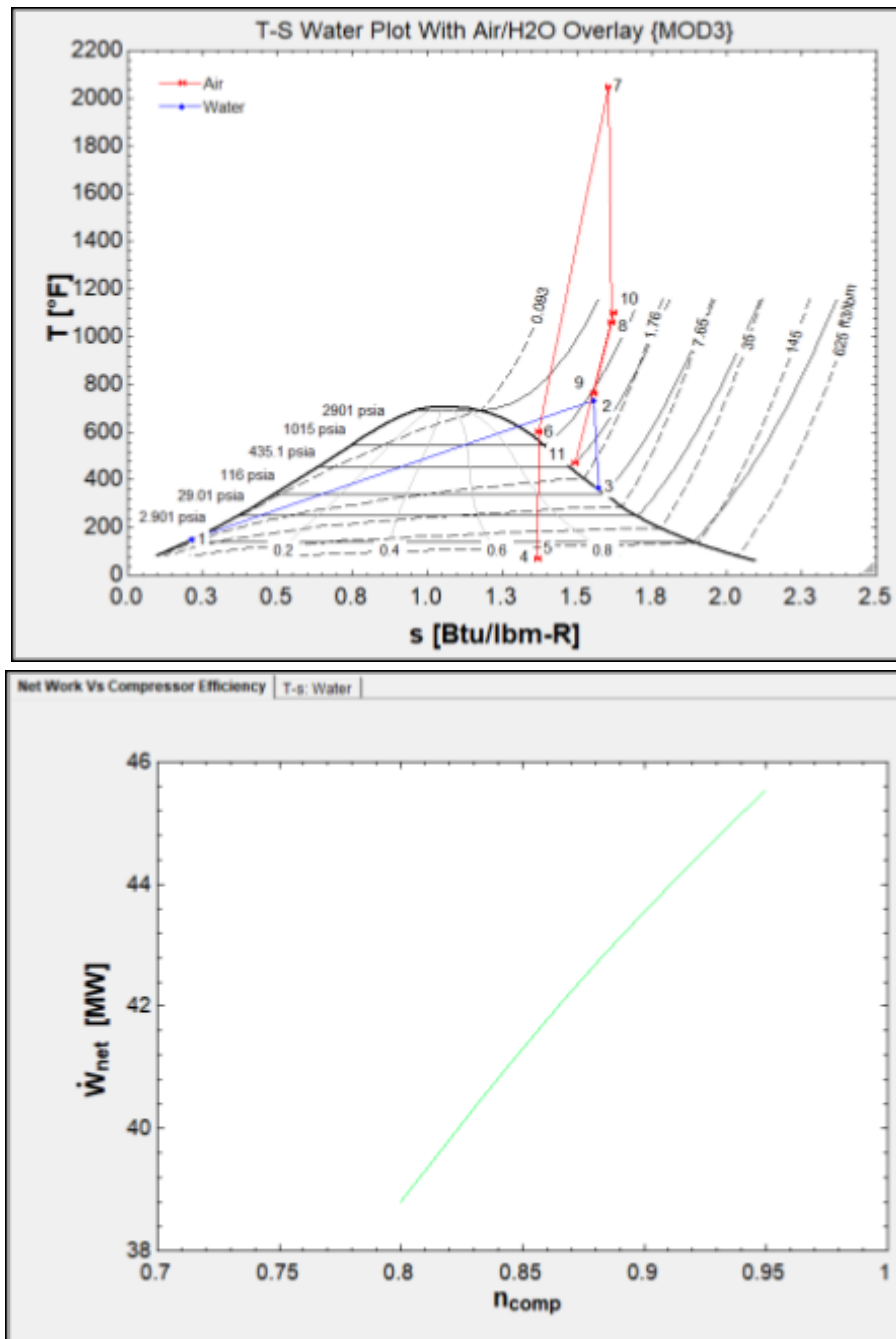
{MOD 2 CHANGE: Calculates the new upgraded MW}
 $\dot{W}_{dot_GT} = \dot{m}_{dot_air} * (h_7 - h_8)$
 $\dot{W}_{dot_GT_MW} = \dot{W}_{dot_GT} * \text{convert}(\text{Btu/s}, \text{MW})$

"WORK CONVS AND MDOT CALCS"

{MOD 2 CHANGE: Fixing $\dot{m}_{dot_air}$ to the baseline value to show power increase}
 $\dot{W}_{dot_raw_GAS} = \dot{W}_{dot_GAS_mw_base} * \text{convert}(\text{MW}, \text{Btu/s})$
 $h_{5_raw} = \text{enthalpy}(\text{Air}, T=69.59)$
 $h_{6_raw} = \text{enthalpy}(\text{Air}, T=674.22)$
 $h_{7_raw} = \text{enthalpy}(\text{Air}, T=2050)$
 $h_{8_raw} = \text{enthalpy}(\text{Air}, T=1034.88)$
 $\dot{m}_{dot_air} = \dot{W}_{dot_raw_GAS} / ((h_{7_raw} - h_{8_raw}) - (h_{6_raw} - h_{5_raw}))$

MOD 3: Finer Air Filter

For our third modification, we modeled a finer inlet air filter before the compressor. This could improve airflow because the air will be cleaner and would reduce long-term compressor blade fatigue. However, the filter also introduced an additional 2 psi pressure drop before the compressor inlet, which creates a trade-off between cleaner airflow and reduced cycle performance [3]. For this analysis, the compressor is assumed to operate at the same pressure ratio as the baseline case while turbine efficiencies remain unchanged. The added inlet pressure loss increased compressor work and negatively affected the overall Brayton cycle performance as shown in EES as the thermal efficiency decreased from 33.13% to 32.62% while the net power output stayed at 42.47 MW. Although the finer filtration can improve certain conditions and reduce compressor fatigue over time, the reduced performance caused by the inlet pressure drop outweighs the benefits [3].



Results and Changes to EES Code:

Unit Settings: Eng F psia mass deg			
$h_1 = 120.2$ [Btu/lb]	$h_{10} = 385.7$ [Btu/lb]	$h_{11} = 224.2$ [Btu/lb]	$h_2 = 1357$ [Btu/lb]
$h_3 = 1200$ [Btu/lb]	$h_{3s} = 1187$ [Btu/lb]	$h_4 = 126.6$ [Btu/lb]	$h_5 = 126.6$ [Btu/lb]
$h_6 = 267.6$ [Btu/lb]	$h_{6s} = 250$ [Btu/lb]	$h_7 = 648.6$ [Btu/lb]	$h_8 = 375$ [Btu/lb]
$h_{8s} = 359.3$ [Btu/lb]	$h_9 = 297.2$ [Btu/lb]	$\dot{m}_1 = 34.34$ [lb/s]	$\dot{m}_{10} = 262.8$ [lb/s]
$\dot{m}_{11} = 262.8$ [lb/s]	$\dot{m}_2 = 34.34$ [lb/s]	$\dot{m}_3 = 34.34$ [lb/s]	$\dot{m}_4 = 262.8$ [lb/s]
$\dot{m}_5 = 262.8$ [lb/s]	$\dot{m}_6 = 262.8$ [lb/s]	$\dot{m}_7 = 262.8$ [lb/s]	$\dot{m}_8 = 262.8$ [lb/s]
$\dot{m}_9 = 262.8$ [lb/s]	$\dot{m}_{air} = 262.8$ [lb/s]	$\dot{m}_{H_2O} = 34.34$ [lb/s]	$\dot{m}_{steam,lb,hr} = 123613$ [lb/hr]
$\eta_{comp} = 0.8753$	$\eta_{cycle} = 0.3262$	$\eta_{tGT} = 0.9457$	$\eta_{tst} = 0.9256$
$P_1 = 905$ [psia]	$P_{10} = 15.7$ [psia]	$P_{11} = 14.7$ [psia]	$P_2 = 855$ [psia]
$P_3 = 156.5$ [psia]	$P_4 = 12.7$ [psia]	$P_5 = 12.7$ [psia]	$P_6 = 137.1$ [psia]
$P_7 = 137.1$ [psia]	$P_8 = 15.7$ [psia]	$P_9 = 15.7$ [psia]	$\dot{Q}_{comb} = 100138$ [Btu/s]
$\dot{Q}_{DB} = 23260$ [Btu/s]	$\dot{Q}_{loss} = 20431$ [Btu/s]	$r_{comp} = 10.8$	$s_1 = 0.2142$ [Btu/lb-R]
$s_{10} = 1.621$ [Btu/lb-R]	$s_{11} = 1.494$ [Btu/lb-R]	$s_2 = 1.557$ [Btu/lb-R]	$s_3 = 1.572$ [Btu/lb-R]
$s_{3s} = 1.557$ [Btu/lb-R]	$s_4 = 1.367$ [Btu/lb-R]	$s_5 = 1.367$ [Btu/lb-R]	$s_6 = 1.383$ [Btu/lb-R]
$s_{6s} = 1.367$ [Btu/lb-R]	$s_7 = 1.604$ [Btu/lb-R]	$s_8 = 1.614$ [Btu/lb-R]	$s_{8s} = 1.604$ [Btu/lb-R]
$s_9 = 1.558$ [Btu/lb-R]	$T_1 = 150$ [F]	$T_{10} = 1102$ [F]	$T_{11} = 471.9$ [F]
$T_2 = 735$ [F]	$T_3 = 368.7$ [F]	$T_4 = 69.59$ [F]	$T_5 = 69.59$ [F]
$T_6 = 646.1$ [F]	$T_7 = 2050$ [F]	$T_8 = 1062$ [F]	$T_9 = 763$ [F]
$\dot{W}_{comp} = 37057$ [Btu/s]	$\dot{W}_{GAS,MW} = 36.78$ [MW]	$\dot{W}_{GT} = 71918$ [Btu/s]	$\dot{W}_{net} = 42.47$ [MW]
$\dot{W}_{ST} = 5397$ [Btu/s]			

Table 1		
	$\dot{W}_{net}$ [MW]	η_{comp}
Run 1	38.79	0.8
Run 2	39.67	0.8167
Run 3	40.5	0.8333
Run 4	41.31	0.85
Run 5	42.08	0.8667
Run 6	42.83	0.8833
Run 7	43.55	0.9
Run 8	44.24	0.9167
Run 9	44.9	0.9333
Run 10	45.55	0.95

"FOUND EFFICIENCIES"

{ $\eta_{comp} = 0.8334$ "Floating this value so comp work vs wnet can be plotted"}
 $\eta_{tGT} = 0.9457$ "MOD"
 $\eta_{tst} = 0.9256$ "MOD"

"State 5 - After Filter"

$T_5 = 69.59$ "T12445 avg"
 $P_5 = (14.7 - 2)$ "Two psi pressure drop from finer filter"

"Baseline compressor pressure ratio"

$r_{comp} = 147.93 / (14.7 - 1)$ "MOD 3: preserve original ratio"

"State 6 - After Compressor"

$P_6 = r_{comp} * P_5$ "MOD 3: maintain same pressure ratio"

"State 6"

$s_{6s} = s_5$
 $h_{6s} = \text{enthalpy}(\text{Air}, s=s_{6s}, P=P_6)$
 $h_6 = h_5 + (h_{6s} - h_5) / \eta_{comp}$ "MOD 3: now depends on unknown efficiency"
 $T_6 = \text{temperature}(\text{Air}, h=h_6)$
 $s_6 = \text{entropy}(\text{Air}, T=T_6, P=P_6)$

"TOTAL WORK AND EFFICIENCY CALCS"

"Find Compressor Efficiency"

$\dot{W}_{dot_GAS_MW} = (\dot{W}_{dot_GT} - \dot{W}_{dot_comp}) * \text{convert}(\text{Btu/s}, \text{MW})$
 $\{\dot{W}_{dot_GAS_MW} = 36.78$ "target to find the required η_{comp} to break even"

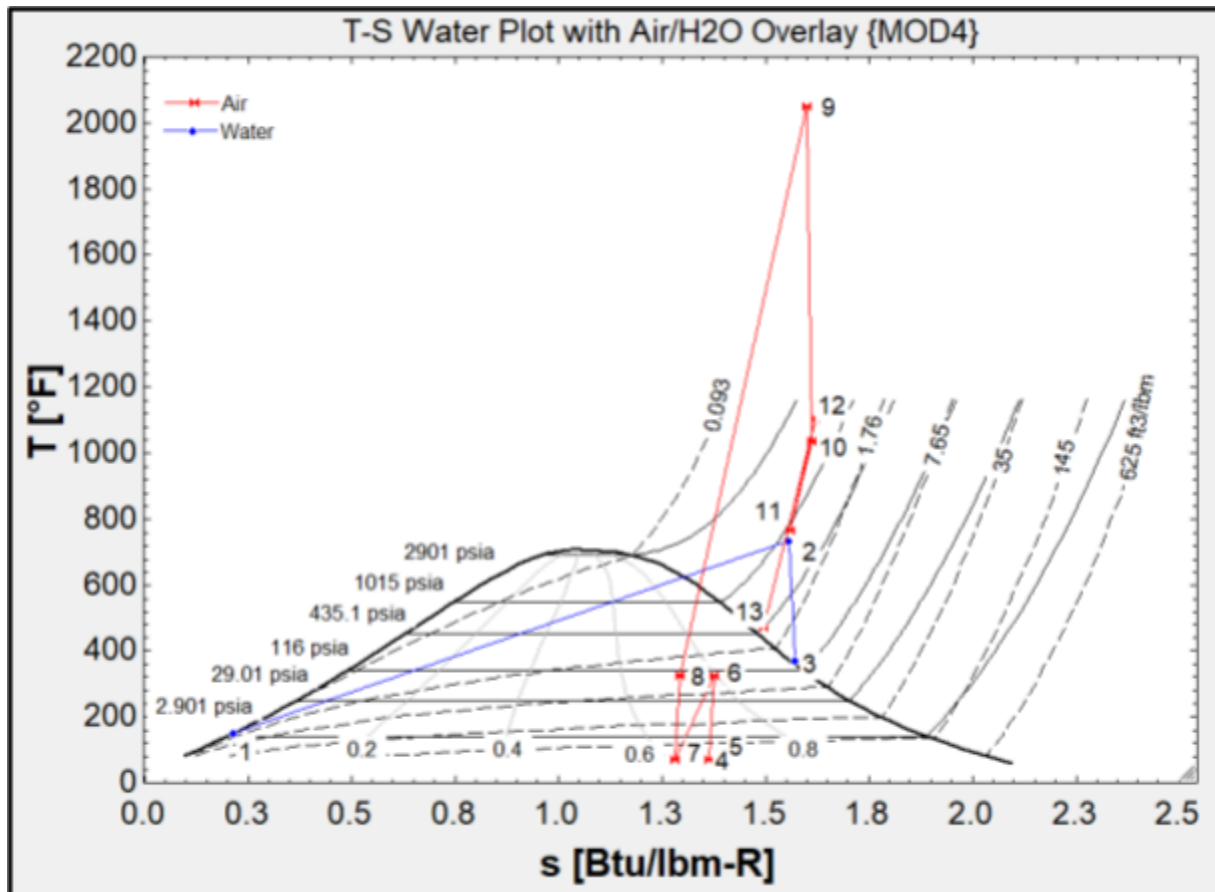
"WORK CONVS AND MDOT CALCS"

" $\dot{M}_{dot}$ - Air Calc"

$\dot{m}_{dot_air} = 262.8$ [lb/s] Found from baseline to see filter impact"

MOD 4: Intercooler Dual Compressor

For our fourth modification, we decided to go with a two-stage compressor with an intercooler. Implementing a dual stage compressor and intercooler greatly reduces the parasitic work taken from the gas turbine. For this modification, the compression process was split into a low-pressure and high pressure compression stage with an intercooler in between. Taken from the textbook the equation $(P_{12}/P_5) = (P_6/P_{13})$ to minimize the required energy for the compressors [4]. Hot air is discharged from the LPC and passes through the intercooler which cools the air back to its original inlet temperature ($T_{13} = T_5$) before it is passed to the inlet of the HPC. Cooling the air increases the density of air molecules while also decreasing the specific volume making it easier for the HPC to compress the air. The air leaving the second compressor will now have the same pressure as in the baseline, but the temperature will be much lower when going to the methane combustor allowing the more air molecules to fuel. From this, the combined compressor work decreases which allows much more GT shaft work to be converted to net electricity for the plant. This is a basic dual stage setup with perfect intercooling assumptions. In real work coolant or “nanofluids” can be used to further increase the performance of the intercooling stage [6].



Results and Changes to EES Code:

Unit Settings: Eng F psia mass deg		
$h_1 = 120.2$ [Btu/lb]	$h_{10} = 385.7$ [Btu/lb]	$h_{11} = 224.2$ [Btu/lb]
$h_{12} = 188.3$ [Btu/lb]	$h_{12a} = 178$ [Btu/lb]	$h_{13} = 126.6$ [Btu/lb]
$h_2 = 1357$ [Btu/lb]	$h_3 = 1200$ [Btu/lb]	$h_{3a} = 1187$ [Btu/lb]
$h_4 = 126.6$ [Btu/lb]	$h_5 = 126.6$ [Btu/lb]	$h_6 = 188.3$ [Btu/lb]
$h_{6a} = 178$ [Btu/lb]	$h_7 = 648.6$ [Btu/lb]	$h_8 = 367.9$ [Btu/lb]
$h_{8a} = 351.7$ [Btu/lb]	$h_9 = 297.2$ [Btu/lb]	$\dot{m}_1 = 34.34$ [lb/s]
$\dot{m}_{10} = 262.8$ [lb/s]	$\dot{m}_{11} = 262.8$ [lb/s]	$\dot{m}_{12} = 262.8$ [lb/s]
$\dot{m}_{13} = 262.8$ [lb/s]	$\dot{m}_2 = 34.34$ [lb/s]	$\dot{m}_3 = 34.34$ [lb/s]
$\dot{m}_4 = 262.8$ [lb/s]	$\dot{m}_5 = 262.8$ [lb/s]	$\dot{m}_6 = 262.8$ [lb/s]
$\dot{m}_7 = 262.8$ [lb/s]	$\dot{m}_8 = 262.8$ [lb/s]	$\dot{m}_9 = 262.8$ [lb/s]
$\dot{m}_{air} = 262.8$ [lb/s]	$\dot{m}_{H_2O} = 34.34$ [lb/s]	$\dot{m}_{steam, b/hr} = 123613$ [lb/hr]
$\eta_{comp} = 0.8334$	$\eta_{cycle} = 0.3242$	$\eta_{LGT} = 0.9457$
$\eta_{tst} = 0.9256$	$P_1 = 905$ [psia]	$P_{10} = 15.7$ [psia]
$P_{11} = 14.7$ [psia]	$P_{12} = 45.02$ [psia]	$P_{13} = 45.02$ [psia]
$P_2 = 855$ [psia]	$P_3 = 156.5$ [psia]	$P_4 = 13.7$ [psia]
$P_5 = 13.7$ [psia]	$P_6 = 147.9$ [psia]	$P_7 = 147.9$ [psia]
$P_8 = 15.7$ [psia]	$P_9 = 15.7$ [psia]	$\dot{Q}_{comb} = 120985$ [Btu/s]
$\dot{Q}_{OB} = 23260$ [Btu/s]	$\dot{Q}_{intercooler, removed} = 16210$ [Btu/s]	$\dot{Q}_{loss} = 18562$ [Btu/s]
$s_1 = 0.2142$ [Btu/lb-R]	$s_{10} = 1.621$ [Btu/lb-R]	$s_{11} = 1.494$ [Btu/lb-R]
$s_{12} = 1.375$ [Btu/lb-R]	$s_{12a} = 1.362$ [Btu/lb-R]	$s_{13} = 1.28$ [Btu/lb-R]
$s_2 = 1.557$ [Btu/lb-R]	$s_3 = 1.572$ [Btu/lb-R]	$s_{3a} = 1.557$ [Btu/lb-R]
$s_4 = 1.362$ [Btu/lb-R]	$s_5 = 1.362$ [Btu/lb-R]	$s_6 = 1.294$ [Btu/lb-R]
$s_{6a} = 1.28$ [Btu/lb-R]	$s_7 = 1.599$ [Btu/lb-R]	$s_8 = 1.61$ [Btu/lb-R]
$s_{8a} = 1.599$ [Btu/lb-R]	$s_9 = 1.558$ [Btu/lb-R]	$T_1 = 150$ [F]
$T_{10} = 1102$ [F]	$T_{11} = 471.9$ [F]	$T_{12} = 325.3$ [F]
$T_{13} = 69.59$ [F]	$T_2 = 735$ [F]	$T_3 = 368.7$ [F]
$T_4 = 69.59$ [F]	$T_5 = 69.59$ [F]	$T_6 = 325.3$ [F]
$T_7 = 2050$ [F]	$T_8 = 1035$ [F]	$T_9 = 763$ [F]
$\dot{W}_{comp} = 32420$ [Btu/s]	$\dot{W}_{GAS, Bu} = 41367$ [Btu/s]	$\dot{W}_{GAS, me} = 43.64$ [MW]
$\dot{W}_{GT} = 73787$ [Btu/s]	$\dot{W}_{HPC} = 16210$ [Btu/s]	$\dot{W}_{LPC} = 16210$ [Btu/s]
$\dot{W}_{net} = 49.34$ [MW]	$\dot{W}_{ST} = 5397$ [Btu/s]	

"Isentropic Compressor and Turbine Assumptions"
 $s_5 = s_{12s}$ {MOD 4}
 $s_{13} = s_{6s}$ {MOD 4}
 $s_7 = s_{8s}$
 $s_2 = s_{3s}$

$\dot{m}_{dot_12} = \dot{m}_{dot_air}$ {MOD 4}
 $\dot{m}_{dot_13} = \dot{m}_{dot_air}$ {MOD 4}

"State 5 - After Filter (ENTERS COMP_1)" {MOD 4}
 $T_5 = 69.59$ "T12445 avg"
 $P_5 = (14.7-1)$ "atm -1"

"INTERCOOLER MODS"

"State 12 - Leaving Comp 1 and Entering IC" {MOD 4}
 $(P_{12}/P_5) = (P_6/P_{13})$ "Eqn from textbook"

"State 13 - Leaving IC and Entering Comp 2" {MOD 4}
 "Assuming perfect intercooling and zero pressure drop"
 $P_{13} = P_{12}$
 $T_{13} = T_5$

"State 6 - After Compressor (LEAVES COMP_2)" {MOD 4}
 $\{T_6 = 674.22$ "T12453 avg"} "MOD"
 $P_6 = 147.93$ "P12451"

```

"State 12 - COMP_1 (LP)" {MOD 4}
s_12s = s_5
h_12s = enthalpy(Air, s=s_12s, P=P_12)
h_12 = h_5 + (h_12s - h_5) / n_comp "Actual state from assumed efficiency"
T_12 = temperature(Air, h=h_12)
s_12 = entropy(Air, T=T_12, P=P_12)

"State 13 - IC Exit" {MOD 4}
h_13=enthalpy(Air, T=T_13)
s_13=entropy(Air, T=T_13, P=P_13)

"State 6 - COMP_2 (HP)" {MOD 4}
s_6s = s_13
h_6s = enthalpy(Air, s=s_6s, P=P_6)
h_6 = h_13 + (h_6s - h_13) / n_comp "Actual state from assumed efficiency and cooled state of 13"
T_6 = temperature(Air, h=h_6) "Finds new Temp"
s_6 = entropy(Air, T=T_6, P=P_6)

```

```

"Air Compressor TWO COMPS" {MOD 4}
W_dot_LPC = m_dot_air * (h_12 - h_5) "Work of Comp 1"
W_dot_HPC = m_dot_air * (h_6 - h_13) "Work of Comp 2"
W_dot_comp = W_dot_LPC + W_dot_HPC "Total parasitic load on GT shaft"

"Intercooler" {MOD 4}
Q_dot_intercooler_removed = m_dot_air * (h_12 - h_13)

"Air Compressor"
{W_dot_comp = m_dot_air * (h_6 - h_5)} {MOD4}

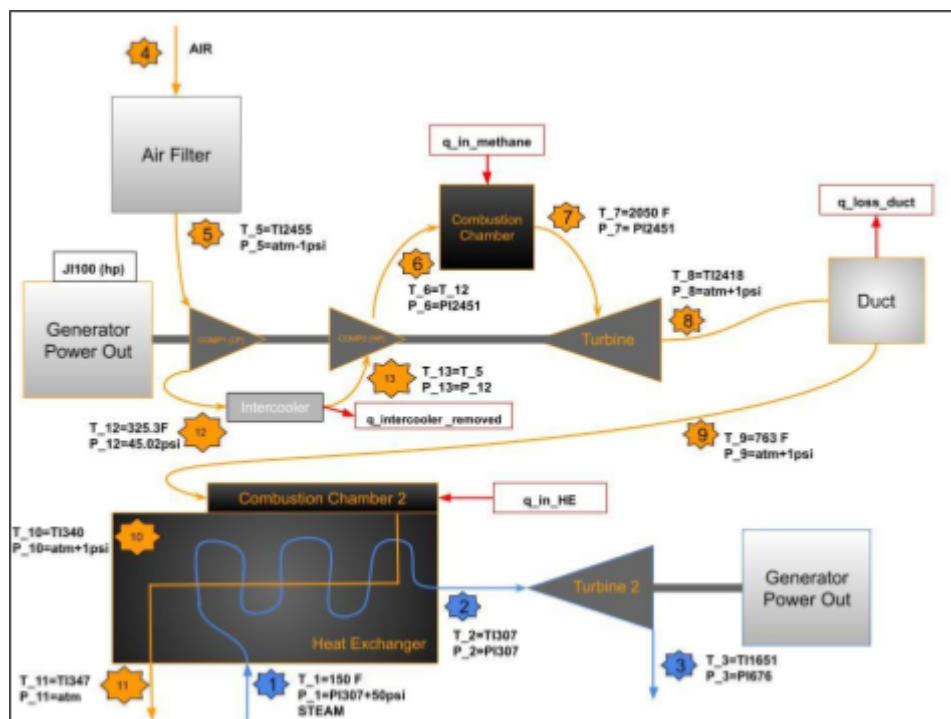
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"WORK CONVS AND MDOT CALCS"
"M_dot - Air Calc - WITH 2 COMPS" {MOD 4}
m_dot_air = 262.8 "[lb/s] FOUND from Baseline"

"Calculate FLOATED Net GT Power"
W_dot_GAS_Btu = m_dot_air * ((h_7 - h_8) - ((h_12 - h_5) + (h_6 - h_13)))
W_dot_GAS_mw = W_dot_GAS_Btu * convert(Btu/s, MW)

```



Comparison Chart:

Unit Settings: SI C kPa kJ mass deg								
dEta _{mod1} = 0.0151	dEta _{mod2} = 0.0044	dEta _{mod3} = -0.0051	dEta _{mod4} = -0.0071	dW _{mod1} = 1.26 [MW]	dW _{mod2} = 1.91 [MW]	dW _{mod3} = 0 [MW]	dW _{mod4} = 6.87 [MW]	
η _{base} = 0.3313	η _{mod1} = 0.3464	η _{mod2} = 0.3357	η _{mod3} = 0.3262	η _{mod4} = 0.3242	pct _{η, mod1} = 4.558 [%]	pct _{η, mod2} = 1.328 [%]	pct _{η, mod3} = -1.539 [%]	
pct _{η, mod4} = -2.143 [%]	pct _{W, mod1} = 2.967 [%]	pct _{W, mod2} = 4.497 [%]	pct _{W, mod3} = 0 [%]	pct _{W, mod4} = 16.18 [%]	W _{base} = 42.47	W _{mod1} = 43.73	W _{mod2} = 44.38	
W _{mod3} = 42.47	W _{mod4} = 49.34							

"MANUAL COMPARISON OF ALL CASES"

"Baseline"

$W_{base} = 42.47$ "[MW]"

$\eta_{base} = 0.3313$

"MOD 1 - Evaporative Cooler"

$W_{mod1} = 43.73$ "[MW]"

$\eta_{mod1} = 0.3464$

"MOD 2 - Turbine Upgrade"

$W_{mod2} = 44.38$ "[MW]"

$\eta_{mod2} = 0.3357$

"MOD 3 - Filter Upgrade"

$W_{mod3} = 42.47$ "[MW]"

$\eta_{mod3} = 0.3262$

"MOD 4 - Dual Stage Compressor With Intercooling"

$W_{mod4} = 49.34$ "[MW]"

$\eta_{mod4} = 0.3242$

"POWER DIFFERENCES"

$dW_{mod1} = W_{mod1} - W_{base}$

$dW_{mod2} = W_{mod2} - W_{base}$

$dW_{mod3} = W_{mod3} - W_{base}$

$dW_{mod4} = W_{mod4} - W_{base}$

"EFFICIENCY DIFFERENCES"

$d\eta_{mod1} = \eta_{mod1} - \eta_{base}$

$d\eta_{mod2} = \eta_{mod2} - \eta_{base}$

$d\eta_{mod3} = \eta_{mod3} - \eta_{base}$

$d\eta_{mod4} = \eta_{mod4} - \eta_{base}$

"PERCENT CHANGES"

$pct_{W_{mod1}} = (dW_{mod1} / W_{base}) * 100$

$pct_{W_{mod2}} = (dW_{mod2} / W_{base}) * 100$

$pct_{W_{mod3}} = (dW_{mod3} / W_{base}) * 100$

$pct_{W_{mod4}} = (dW_{mod4} / W_{base}) * 100$

$pct_{\eta_{mod1}} = (d\eta_{mod1} / \eta_{base}) * 100$

$pct_{\eta_{mod2}} = (d\eta_{mod2} / \eta_{base}) * 100$

$pct_{\eta_{mod3}} = (d\eta_{mod3} / \eta_{base}) * 100$

$pct_{\eta_{mod4}} = (d\eta_{mod4} / \eta_{base}) * 100$

Discussion

The EES analysis shows that although the baseline plant already operates fairly efficiently, these upgrades can be beneficial to system performance. The baseline plant operated at a thermal efficiency of 33.13% and produced a net power output of 42.47 MW. Each modification affected the cycle differently, creating trade-offs between efficiency, power generation, complexity, and component lifespan.

The evaporative cooling system produced the largest increase in thermal efficiency. The efficiency increased from 33.13% to 34.64%, which was an increase of 4.56%. Net power output also increased to 43.73 MW, representing a gain of 1.26 MW or 2.97%. These improvements happened because lowering the compressor inlet temperature increased air density and reduced compressor work. While this modification produced the best efficiency improvement, the overall power increase was smaller than some of the other upgrades, and other factors like the ROI create more negative examples for this mod.

The OEM turbine upgrade provided the best overall results based on how much it increased efficiency and power output. Increasing the turbine inlet temperature to 2100 °F increased thermal efficiency from 33.13% to 33.57%, which was a gain of 1.31%, while net power output increased from 42.47 MW to 44.38 MW, which was a gain of 1.91 MW or 4.50%. This modification made improvements to gas turbine performance and HRSG effectiveness, which makes it a very competitive upgrade option for Chevron.

The high-efficiency inlet air filter produced more negative results than the other options. The added 2 psi pressure drop reduced thermal efficiency from 33.13% to 32.62% which was a decrease of 1.54% while the net power output stayed at 42.47 MW. This made the modification the worst option because of the economic and operational factors.

The dual-stage compressor with intercooling created the largest increase in net power. The net power output increased from 42.47 MW to 49.34 MW which was a gain of 6.87 MW or 16.18%. However the thermal efficiency decreased from 33.13% to 32.42% which was a decrease of 2.14%. This modification made improvements, larger than the OEM modification, which makes it a very strong option for Chevron, but there was also the trade-off of the loss of thermal efficiency.

Conclusions

This project successfully modeled and analyzed a Chevron combined-cycle power plant using EES to calculate the improvements of multiple plant upgrades and show their potential effects on thermal efficiency and net power generation.

The baseline plant operated at a thermal efficiency of 33.13% and generated 42.47 MW of net power. Among the four modifications analyzed, the evaporative cooling system produced the highest efficiency increase, improving efficiency to 34.64% which was a gain of 4.56%. The

OEM upgrade gave a balanced upgrade in both efficiency and power generation. Increasing the turbine inlet temperature to 2100 °F increased efficiency to 33.57% and increased net power output to 44.38 MW which was a 4.50% increase in power generation. Because this upgrade improved both Brayton cycle performance and HRSG effectiveness, it represents the strongest overall recommendation for Chevron. The high-efficiency inlet air filter gave the weakest results of the 4 mods. The efficiency decreased to 32.62% and net power output stayed at 42.47 MW. However, it should be observed that with the air filter varying the efficiency of the compressor does have a positive effect on net work if it is greater than the baseline as shown in the graph under mod 2. The dual-stage compressor with intercooling produced the largest increase in net power output which increased generation to 49.34 MW which was a gain of approximately 16.18%. However, overall efficiency decreased to 32.42% making a trade-off between the increase in power output and loss in thermal efficiency. Additionally, the added complexity and cost of implementing an additional compressor should be considered for the feasibility of this modification.

Overall, the analysis shows us that each modification improves a different aspect of the plant's performance. The evaporative cooler produced the greatest efficiency improvement, the intercooling system generated the largest increase in power output, and the OEM turbine upgrade provided an overall combination of increased efficiency and power generation. The intercooler provided a very large increase in net power output but came with a trade-off of decreased thermal efficiency. With all of the factors provided in our analysis, we will be recommending the implementation of the OEM turbine upgrades to Chevron. This suggestion is based on the all-around benefits of implementing this modification to the base cycle. The balance of increases in both power and efficiency outweigh the results of the intercooler, as even though the intercooler provided a very large increase in power, the complexity of adding this component combined with potential increased fuel usage and other negative effects of a lower thermal efficiency is not enough to justify going with this modification. The OEM upgrades provide an option that positively affects both production and consumption of the cycle, while also being a much simpler upgrade to the turbines instead of trying to implement complete new component assemblies to the cycle. Chevron should implement the OEM modifications to the turbine parts for a well-rounded upgrade to the system and cycle as a whole.

References

MOD 1 Source:

[1] O. Zeitoun, M. Ali, and H. Al-Ansary, "Two-Stage Evaporative Inlet Air Gas Turbine Cooling," *Energies*, vol. 14, no. 5, Art. no. 1382, Mar. 2021, doi: 10.3390/en14051382. [Online]. Available: [[Two-Stage Evaporative Inlet Air Gas Turbine Cooling](#)]

MOD 2 Source:

[2] A. M. Elkady, D. M. Kalitan, J. Herbon, G. Leonard, R. Akula, H. Karim, and M. Hadley, "Gas Turbine Emission Characteristics in Perfectly Premixed Combustion," in *Proc. ASME Turbo Expo: Turbine Technical Conference and Exposition*, Vancouver, BC, Canada, 2011, Paper GT2011-46470, doi: 10.1115/GT2011-46470. [Online]. Available: [[Gas Turbine Emission Characteristics in Perfectly Premixed Combustion | GT | ASME Digital Collection](#)]

MOD 3 Source:

[3] A. O. Büyükköse and A. Aslan, "Machine Learning Analysis of Inlet Air Filter Differential Pressure Effects on Gas Turbine Power and Efficiency with Carbon Footprint Assessment," *Machines*, vol. 14, no. 2, Art. no. 170, Feb. 2026, doi: 10.3390/machines14020170. [Online]. Available: [[Machine Learning Analysis of Inlet Air Filter Differential Pressure Effects on Gas Turbine Power and Efficiency with Carbon Footprint Assessment](#)]

MOD 4 Sources:

[4] Y. A. Çengel, M. A. Boles, and M. Kanoğlu, "The Brayton cycle with intercooling, reheating, and regeneration," in *Thermodynamics: An Engineering Approach*, 10th ed. New York, NY, USA: McGraw-Hill Education, 2024, sec. 10-10.

[5] L. Xu, X. Niu, W. Hong, and W. Su, "A comprehensive multi-objective optimization study on the thermodynamic performance of a supercritical CO₂ Brayton cycle incorporating multi-stage main compressor intermediate cooling," *Energies*, vol. 17, no. 24, p. 6372, 2024. <https://doi.org/10.3390/en17246372>

[6] A. Alsayegh and N. Ali, "Gas turbine intercoolers: introducing nanofluids—a mini-review," *Processes*, vol. 8, no. 12, p. 1572, 2020. <https://doi.org/10.3390/pr8121572>